

Reviewer Pack — Minimal Rank of Tropical Logarithmic Forms vs Resolution Pro...

Entry 2161bbe9ad19 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tropical Logarithmic Forms vs Resolution Proof Length for Tseitin Formulas

Entry ID: 2161bbe9ad19 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 21:24:00 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Logarithmic Forms) **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity for Tseitin Formulas

Statement:

For every Tseitin formula F on an expander graph G , the resolution proof length for F is $\Omega(2^{\text{rank}(T(F))})$, where $\text{rank}(T(F))$ is the minimal rank of a tropical logarithmic form associated with the characteristic polynomial of the Tutte polynomial $T(G)$.

Rationale (proposer's reasoning):

Tropical logarithmic forms are expected to encode complex algebraic properties that arise in the analysis of expander graphs. The Tutte polynomial is a fundamental invariant for graphs, and its characteristic polynomial has been shown to relate to resolution complexity. This conjecture suggests that the rank of this form can provide a lower bound on resolution proof length.

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e56a3afd370267e6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given expander graph G with $n \leq 40$ vertices, if the average ratio of the resolution proof length to $2^{\text{rank}(T(F))}$ across all seeds is less than or equal to 1.5, then it supports the conjecture; otherwise, it falsifies the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	1.00	SAFE	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical geometry AND logarithmic forms AND resolution proof complexity - Tseitin formulas AND Tutte polynomial AND tropical logarithmic forms AND minimal rank - resolution proof length expander graphs AND tropical geometry minimal rank

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        if matrix[i][i] == 0:
            return None # Singular matrix, no unique solution
        for j in range(i + 1, n):
            factor = -matrix[j][i] / matrix[i][i]
            for k in range(n):
                matrix[j][k] += factor * matrix[i][k]
    rank = sum(1 for row in matrix if any(row))
    return rank

def tseitin_formula(n):
    variables = list(range(1, n + 1))
    clauses = []
    for i in range(n - 1):
        clauses.append([variables[i], variables[i + 1]])
    clauses.append([-variables[-2], variables[-1]])
    clauses.append([variables[0]])
    return clauses

def resolution_length(clauses):
    # Simplified version of DPLL algorithm to estimate resolution length
    stack = []
    while True:
        unit_clause = next((c for c in clauses if len(c) == 1), None)
        if not unit_clause:
            break
        literal = unit_clause[0]
```

```

        clauses = [c for c in clauses if literal not in c and -literal not in c]
        stack.append(literal)
    return len(stack)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    G = tseitin_formula(n)
    rank = gaussian_elimination(G)
    if rank is None:
        return {
            "metric_name": "Rank vs DPLL Heig",
            "metric_value": float('inf'),
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": "Singular matrix"
        }

    resolution_len = resolution_length(G)
    ratio = resolution_len / (2 ** rank)
    return {
        "metric_name": "Rank vs DPLL Heig",
        "metric_value": ratio,
        "instances_tested": 1,
        "conjecture_holds": ratio <= 1.5,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    metric_values = [r["metric_value"] for r in results if r["instances_tested"] > 0]
    conjecture_holds = all(r["conjecture_holds"] for r in results if r["instances_tested"] > 0)

    if conjecture_holds:
        mean = sum(metric_values) / len(metric_values)
        std = math.sqrt(sum((x - mean)**2 for x in metric_values) / len(metric_values))
        support_fraction = len(results) / len(seeds)
        print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results if r["instances_tested"] > 0):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"First failing seed\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE no seeds supported the conjecture")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_87ebec1c.py", line 81, in <module>
    trial_result = run_trial(seed)
                   ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_87ebec1c.py", line 55, in run_trial
    rank = gaussian_elimination(G)
           ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_87ebec1c.py", line 26, in gaussian_elim
    matrix[j][k] += factor * matrix[i][k]
    ~~~~~^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which means that it was unable to complete the computation required to evaluate the conjecture. | next: Re-run the test with appropriate error handling to ensure it completes without crashing and produces the necessary data for evaluation.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11308
2	propose	ollama_remote	glm4:latest	0	0	9651
3	preregistration	ollama_remote	glm4:latest	0	0	5292
4	novelty	ollama_remote	glm4:latest	0	0	4629
5	novelty	ollama_remote	glm4:latest	0	0	5317
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29946
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12659
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13302
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9411
10	judge	ollama_remote	glm4:latest	0	0	9144

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 110659 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/2161bbe9ad19.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/2161bbe9ad19.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/2161bbe9ad19.tar.gz` (if generated)